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A data-driven distributed force model for wall-modeled large-eddy simulations of rough-wall turbulence

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ABSTRACT

In this work, a data-driven distributed force model for wall-modeled large-eddy simulations of rough-wall turbulence is proposed. In the proposed model, a body force with its distribution given by the area fraction of solid is distributed in the near-wall roughness layer to model the impact of the rough wall on the outer flow. The amplitude of the body force is computed based on the local streamwise velocity and the equivalent sandgrain roughness height, with the latter calculated using a deep neural network (DNN) model. Ideally, the proposed model can be applied to any type of rough wall if the geometrical parameters required by the DNN model are available. The proposed model is tested by simulating the turbulent channel flow for three cube-roughened surfaces and three ellipsoid-roughened surfaces at different grid resolutions and different Reynolds numbers. An overall good agreement with the direct numerical simulation results is achieved.

1. Introduction

Rough wall turbulence widely exists in environmental and engineering applications [1,2]. The scale of roughness elements is often orders of magnitude smaller than the scale of the flows of interest [3]. Carrying out roughness-resolved simulations of rough-wall turbulent flows face more challenges in comparison with the smooth-wall counterpart, that the grid needs to be refined in both wall-normal and wall-tangential directions in the near-wall region, adding extra computational cost. An alternative way is to model the effects of the near-wall region on the outer flow, e.g., the so-called wall-modeled large-eddy simulation (WMLES or wall-modeled LES), to avoid the need of resolving the near-wall small-scale turbulence [1,4–7]. The major challenge for wall-modeled LES is the lack of a universal wall model for various flow regimes and various surfaces. In this work, we develop a wall model for rough-wall turbulence, which employs a distributed force to represent the effects of the surface roughness and trains a data-driven model to compute the roughness length of the surface, respectively, and then test the model using the data from roughness-resolved simulations.

In wall-modeled LES, the effects of the near-wall unresolved flow on the outer flow are modeled via approximate boundary conditions [8–10]. The shear stress boundary condition on the wall is the often employed approximate boundary condition. For a smooth wall, the wall shear stress can be computed via the law of the wall using the tangential velocity at the near-wall grid points (often the first off-wall grid point). To compute the wall shear stress for a rough wall, the roughness length is required additionally.

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In the literature, many works were devoted to establishing relations between the roughness length and the topography of the rough surface, without achieving a universal model working for various rough surfaces [2]. The roughness can be classified as k-roughness and d-roughness [11]. For k-roughness, k_s is proportional to the topography parameters of the rough surface. For instance, the normalized k_s increases with the solidity in the dense regime, while decreasing with the solidity in the sparse regime. The often employed roughness topography parameters include the averaged roughness height k_{avg}, the second-order, third-order and fourth-order moments of roughness height fluctuations (i.e., k_{rms}, S_k and K_u), the effective slope E_s, the porosity P_o and others [2]. Models for correlating the equivalent sandgrain roughness height k_s and the above roughness topography parameters have been developed in the literature either using an empirical approach [12–15], a theoretical approach [16,17], or a data-driven approach [18–20]. For instance, the model developed by Forooghi et al. [21] relates k_s to the peak-to-trough height k_c, the S_k and the effective slope E_s for rough surfaces formed by columnars of various shapes. Barros et al. [22] and Flack et al. [23] formulated the expression for k_s using k_{rms} and S_k for wavy surfaces with low E_s. Applying certain physical constraints are beneficial for deriving the k_s models. The assumptions of the logarithmic law for velocity and the constant shear stress were employed by Macdonald et al. [16]. The roughness-element-scale flow sheltering together with the momentum and mass conservation was considered in the model proposed by Yang et al. [24]. The major challenge for developing a k_s model is to ensure its generalizability. The machine learning method provides a new approach for approximating k_s as a function of topography parameters. In the work by Jouybari et al. [18], a deep neural network (DNN) model and a Gaussian process regression model were trained for computing k_s using the data for 45 rough surfaces. In the work by Yang et al. [19], the model were improved by retraining the model using a larger dataset of 68 homogeneous rough surfaces. In the recent work by Ma et al. [20], a particle swarm optimized backpropagation method is proposed to predict k_s using the data from DNS, LES and experimental cases for 86 rough surfaces.

Employing a distributed force to approximate the effect of rough wall is considered to be more suitable than a shear stress boundary condition in roughness-modeled eddy-resolving simulations of rough-wall turbulence [25]. Durbin and Varghese [25] showed that the distributed force approach is capable of generating large eddies in the outer region. The approach based on the shear stress boundary condition, on the other hand, may underpredict the size of eddies in outer layer, and cannot accurately predict the peaks of the Reynolds stress terms [26]. A distributed force has been employed to model unresolved dynamics in the literature. For instance, a distributed force was employed to model plant canopies in LES of atmospheric surface layer [27]. In Reynolds-Averaged Navier-Stokes (RANS) simulation of rough-wall turbulence, the so-called discrete element model employs distributed forces to represent the effects of rough surfaces on momentum and heat transfer [28]. In the eddy-resolving simulations of rough-wall turbulent flows, several works have been carried out in the literature. In the model proposed by Cui et al. [29], the topography of rough surfaces is decomposed into the resolved scale and the subgrid scale (analogous to the idea of dynamic subgrid-scale model), in which the former is modeled in a way similar to the immersed boundary method and the latter is modeled by a randomly distributed body force, respectively, with the model tested using the case of flow in a wavy channel. The model by Anderson and Meneveau [30] is proposed for the flow over slow-varying terrains, that the rough surface is resolved in the horizontal directions but not in the vertical direction. In their model, a form drag, which is expressed by the frontal area and the incoming momentum flux of the flow, is proposed to capture the effect of vertically unresolved rough surface. Busse and Sandham [31] proposed to employ a roughness factor, the local velocity and a shape function to model the influence of rough wall on the outer flows, with the former two and latter one determining the amplitude and distribution of the body force, respectively. In Busse and Sandham's model, the body force is quadratic to the local velocity. In the modified model by Forooghi et al. [32], a forcing term linear to the local velocity was introduced to represent the viscous drag due to roughness. In the model proposed by Krumbein et al. [33] for very large-eddy simulation, a shape function was also employed to consider the geometry of roughness element. Even with the above advances, the major challenge remains, which is how to relate the model to real-life rough surfaces. This includes all stages of the model development, i.e., deviation, testing and applications.

In this work, we propose to combine a distributed force model and a data-driven equivalent sandgrain roughness height (k_s) model for roughness-modeled large-eddy simulation of rough-wall turbulence. In the proposed approach, the force distribution is determined by the vertical distribution of the area fraction of roughness elements, and the force amplitude is determined by the data-driven k_s model. Employing several main geometric statistics of the rough surfaces for determining the amplitude and distributions of the forcing term, which can directly be related to a specific type of rough surfaces, is the key contribution of this work. The proposed approach is tested using the turbulent channel flow cases for various rough surfaces for different grid resolutions and Reynolds numbers.

In the rest of the paper, the numerical methods for flow simulations, the proposed distributed force model, and the data-driven k_s model are first described in Section 2. The test results for the roughness-modeled large-eddy simulation approach are then presented in Section 3. Lastly, conclusions from this work are drawn in Section 4.

2. Methodology

2.1. Flow solver

We employ the Virtual Flow Simulator (VFS-Wind) [34–37] code for the simulation of turbulent flows over rough walls. In the VFS-Wind code, the governing equations are the three-dimensional unsteady spatially filtered incompressible Navier-Stokes equations in Cartesian coordinates shown as follows:

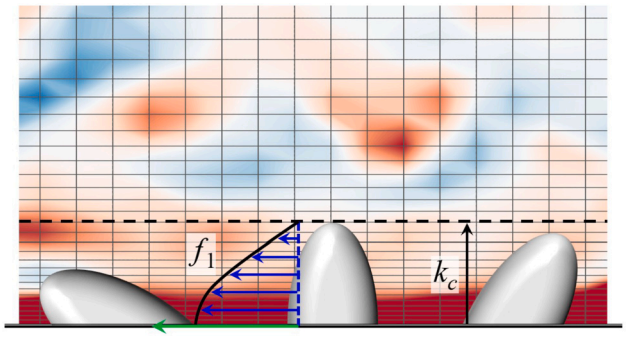

Fig. 1. Schematic of the distributed force model in the near-wall region.

$$\begin{cases} \dfrac{\partial u_j}{\partial x_j} = 0, \\ \dfrac{\partial u_i}{\partial t} = -\dfrac{\partial u_i u_j}{\partial x_j} - \dfrac{1}{\rho}\dfrac{\partial p}{\partial x_i} + \nu \dfrac{\partial^2 u_i}{\partial x_j \partial x_j} - \dfrac{\partial \tau_{ij}}{\partial x_j} + f_i, \end{cases} \tag{1}$$

where x_i ($i = 1$, 2, and 3) are the Cartesian coordinates, u_i is the i-th component of the filtered velocity vector, ρ, ν and p denote the fluid density, kinematic viscosity and pressure, respectively, and τ_{ij} represents the anisotropic part of the subgrid-scale stress tensor. The τ_{ij} is modeled by the Smagorinsky model,

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\nu_t \overline{S}_{ij}, \tag{2}$$

where $\overline{S}_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$ is the filtered strain-rate tensor and ν_t is the eddy viscosity calculated by

$$\nu_t = C\Delta^2 |\overline{S}|, \tag{3}$$

where C is the model coefficient calculated dynamically using the procedure of Germano et al. [38], $|\overline{S}| = \sqrt{2\overline{S}_{ij}\overline{S}_{ij}}$, and Δ is the filter size. The forcing term f_i models the effects of the rough surface on the flow.

The governing equations are discretized in space using a second-order accurate central differencing scheme, and integrated in time using the fractional step method. An algebraic multigrid acceleration along with generalized minimal residual method (GMRES) solver is used to solve the pressure Poisson equation. A matrix-free Newton-Krylov method is used for solving the discretized momentum equation.

2.2. Distributed force model

In this study, the influence of the rough surface on the outer flow is represented using a vertical-distributed force. It is a reasonable approximation. Although the effect of the rough wall is considered as a friction drag from the perspective of the outer flow, the rough surface actually acts on the flow in terms of both friction drag and form drag in the near-surface layer occupied by the roughness elements (Fig. 1). The expression of the distributed force (f_i in Eq. (1)) is given as follows,

$$f_i(x, y, z) = -c_i \alpha_f |u_i(x, k_c, z)| u_i(x, k_c, z), \tag{4}$$

where c_i ($i = 1, 2, 3$) are the force coefficients, α_f describes the force distribution in the vertical direction, and $u_i(x, k_c, z)$ are the instantaneous velocity at the upper edge of the roughness layer (i.e., $y = k_c$ the crest of the roughness element), similar to that in Durbin and Varghese [25]. The rationale for Eq. (4) can be found in Stripf et al. [28].

To compute the values of the force coefficients c_i, the vertical force distribution needs to be given. In the proposed approach, α_f is assumed to be the same as the area fraction of roughness elements at various vertical locations, which was also used in RANS and in atmospheric applications [27,39,40]. For the cases considered, the spanwise and vertical force coefficients are zero (i.e., $c_2 = c_3 = 0$). The streamwise force coefficient c_1 is determined using the equivalence between the drags of the rough surface on the outer flow in the wall shear stress form and in the distributed force form as follows,

$$\iiint_{\Omega} f_1 \, dx dy dz = \iint_{S} -\frac{\tau_w}{\rho} dx dz, \tag{5}$$

where Ω denotes the three-dimensional domain of $[0, L_x] \times [0, k_c] \times [0, L_z]$, S denotes the two-dimensional plane of $[0, L_x] \times [0, L_z]$. With the logarithmic law for rough wall, $\frac{U}{u_\tau} = \frac{1}{\kappa}\ln(\frac{y}{k_0})$ [41] (where the roughness length $k_0 = 0.033k_s$ [11], $\rho u_\tau^2 = \tau_w$, and $\kappa = 0.4$ is the Kármán constant), thus the wall shear stress τ_w can be determined by

$$\frac{\tau_w}{\rho} = \kappa^2 \left(\frac{U}{\ln(y/k_0)} \right)^2 . \tag{6}$$

Using the streamwise velocity $u_1(x, y_m, z)$ at $y = y_m$, the wall shear stress can be obtained and the force coefficient c_1 can be expressed as follows,

$$c_1 = \frac{\iint_S \kappa^2 \left(\frac{u_1(x,y_m,z)}{\ln(y_m/k_0)} \right)^2 dxdz}{\iiint_\Omega \alpha_f |u_1(x, y_m, z)| u_1(x, y_m, z) dxdydz}. \tag{7}$$

Introducing $H = \int \alpha_f dy$ and replacing the horizontal integration domain $\int (\cdot) dxdz$ with the local region, a more general form of Eq. (7) can be obtained as follows,

$$c_1 = \mathrm{sgn}(u_1) \frac{\kappa^2}{\langle H(\ln(y_m/k_0))^2 \rangle_s}, \tag{8}$$

where $\mathrm{sgn}(u_1)$ is the sign function, and $\langle \cdot \rangle_s$ denotes the averaging over a local area s. The above expression in principle can be applied to a wall with inhomogeneous roughness distribution, given that the logarithmic law still exists. Another note is that, the force coefficient has been formulated in a way without the zero-plane displacement (d), as a model of d often does not exist for a rough surface. This means that the computational domain in the wall-normal direction essentially starts from d instead of the smooth wall when employing the proposed formulae in WMLES.

2.3. Data-driven k_s model

The equivalent sandgrain roughness height k_s (or the roughness length k_o), which is required for computing τ_w in Eq. (6), is computed using a data-driven k_s model. The output label is $\mathbf{Y} = [k_s]$. The input geometrical parameters ($\mathbf{X} = [S_x, k_{\mathrm{rms}}, S_k, K_u, E_x, P_o]$) include S_x the averaged streamwise element spacing, k_{rms}, S_k, and K_u the root-mean-square (rms), skewness, flatness of roughness height fluctuations, respectively, E_x the effective slope of roughness in the streamwise direction, and P_o the surface porosity. The above geometrical parameters are defined as follows,

$$\begin{cases} k_{\mathrm{avg}} = \dfrac{1}{A_t} \displaystyle\int_{x,z} k(x,z) dA, \\ k_{\mathrm{rms}} = \sqrt{\dfrac{1}{A_t} \displaystyle\int_{x,z} \left(k(x,z) - k_{\mathrm{avg}} \right)^2 dA}, \\ R_a = \dfrac{1}{A_t} \displaystyle\int_{x,z} \left| k(x,z) - k_{\mathrm{avg}} \right| dA, \\ S_k = \dfrac{1}{A_t} \displaystyle\int_{x,z} \dfrac{\left(k(x,z) - k_{\mathrm{avg}} \right)^3}{k_{\mathrm{rms}}^3} dA, \\ K_u = \dfrac{1}{A_t} \displaystyle\int_{x,z} \dfrac{\left(k(x,z) - k_{\mathrm{avg}} \right)^4}{k_{\mathrm{rms}}^4} dA, \end{cases} \tag{9}$$

$$\begin{cases} E_x = \dfrac{1}{A_t} \displaystyle\int_{x,z} \left| \dfrac{\partial k(x,z)}{\partial x} \right| dA, \\ P_o = \dfrac{1}{A_t k_c} \displaystyle\int_0^{k_c} A_p(y) dy, \\ I_x = \tan^{-1} \left[\dfrac{1}{2} S_k \left(\dfrac{\partial k}{\partial x} \right) \right], \end{cases} \tag{10}$$

where A_t is the total plan area, A_p is the plan area of the roughness element, $k(x, z)$ is the height distribution of the rough surface, k_{avg} is the mean roughness height, R_a is the first-order moment of roughness height fluctuations, and I_x is the streamwise inclination angle. It is noted that the number of input features employed in this work is 6, which is significantly less than those employed by [18]. We will show that the employed six input features are adequate for the rough surfaces considered in this work.

The neural network is employed to approximate the relation between the geometrical parameters of the rough surface and the equivalent sandgrain roughness height. The employed feedforward neural network (FNN) consists of an input layer, multiple hidden layers, and an output layer. Each layer has a number of neurons, which are computational units that propagate the weighted sums of the inputs to an activation function and compute the output. The employed activation function is the rectified linear unit (ReLU) [42]. And the loss function is defined as

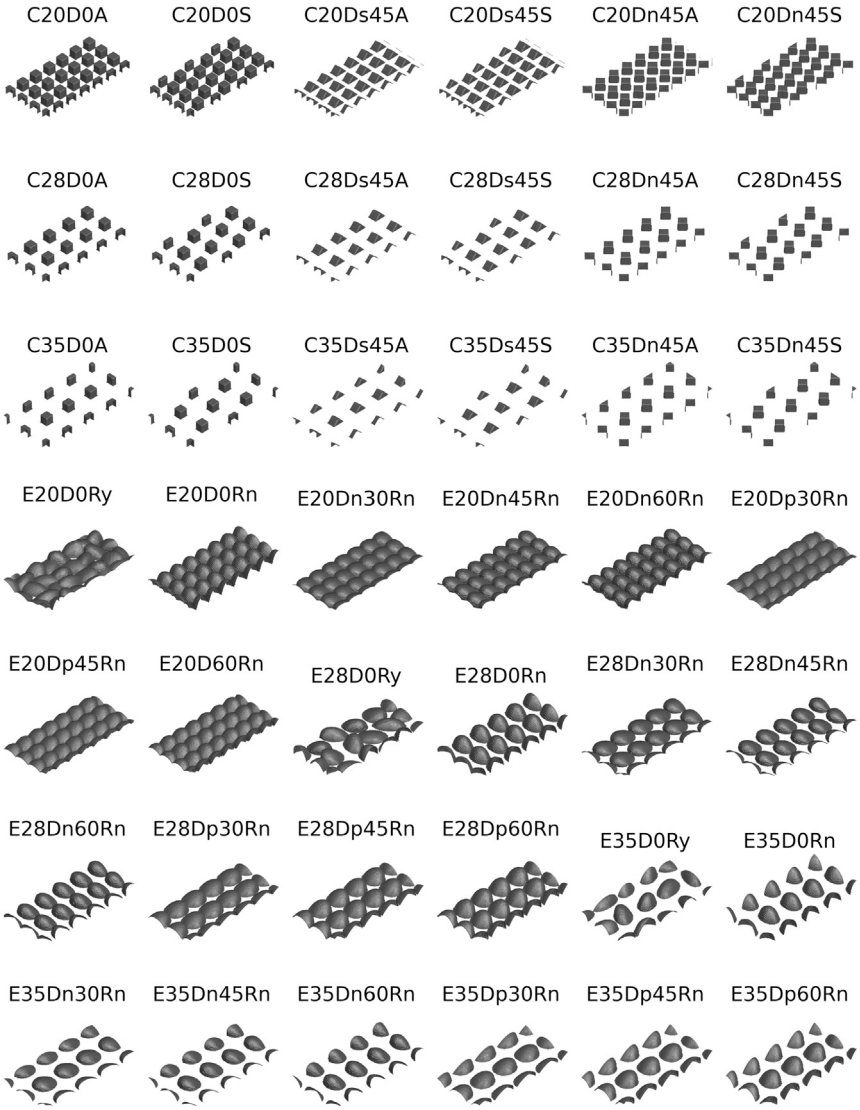

Fig. 2. Schematic of the 42 samples of rough surfaces in the $x-z$ plane from our previous work [45,46], where "C" and "E" denote the cube- and ellipsoid-shaped roughness elements, respectively.

$$E_k = \frac{1}{N_s}\sum_{i=1}^{N_s}\left(\mathbf{Y}_i - \mathbf{Y}_i^{\,*}\right)^2 + \frac{\lambda_0}{2N_s}\,\|w\|_2^2, \tag{11}$$

where N_s is the number of training samples, w is the weight coefficient, and λ_0 is the regularization rate, which is set to 0.001. The first term in Eq. (11) denotes the mean square error (MSE) between the model output $\mathbf{Y}^*$ and the labeled output $\mathbf{Y}$. The second term is an L2 regularization term included to avoid overfitting. We use the error backpropagation (BP) scheme [43] implemented with Scikit-learn [44] to train the FNN model. To optimize the loss function, the LBFGS (Limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm), an optimizer applicable to small dataset based on Quasi-Newton method, is used to adjust the weight and bias coefficients. The learning rate is set to be "adaptive" with the initial value of 0.01.

In the model training, in total 118 samples of various rough surfaces are employed. They include the 44 samples employed by Jouybari et al. [18], the 32 samples from the work by Forooghi et al. [21], and 42 samples from our previous work [45,46]. Fig. 2 plots the geometries of the 42 samples from our previous work, which includes the various ellipsoid- and cube-shaped roughness elements. The details of all the corresponding geometrical parameters and k_s are shown in Table A.5 of Appendix A. In the training of the data-driven k_s model, 70% of the samples are employed as the training dataset with the remaining 30% as the testing dataset. Different combinations of the number of hidden layers and the number of neurons in each hidden layer were tested. The neural network employed in the production runs includes 5 hidden layers with 64, 32, 16, 8, and 4 neurons in sequence, with the maximum error less than 40% and the average error less than 5%. Here, the average error is calculated by averaging the relative error ($100\% \times |k_{s,p} - k_s|/k_s$) based on the predicted $k_{s,p}$ and real k_s. The performance of other network configurations is shown in Table 1.

Table 1
The error of data-driven model training with various neural network configurations. The numbers in "L1"∼"L6" denote the neurons in different hidden layers.

L1	L2	L3	L4	L5	L6	$error_{\max}$ (%)	$error_{\text{avg}}$ (%)
6	6	6	6	6	6	96.3516	10.0044
12	12	12	12	12	12	74.9547	6.9636
24	24	24	24	24	24	44.7949	5.4613
64	32	16	8	4	2	33.5215	5.7495
12	24	48	24	12	6	37.8918	5.7978
64	32	16	8	4	–	37.1046	4.8511
4	8	16	32	64	–	50.3870	11.1246
64	32	16	8	–	–	55.0152	6.0052
128	64	32	16	8	4	42.5700	4.2879

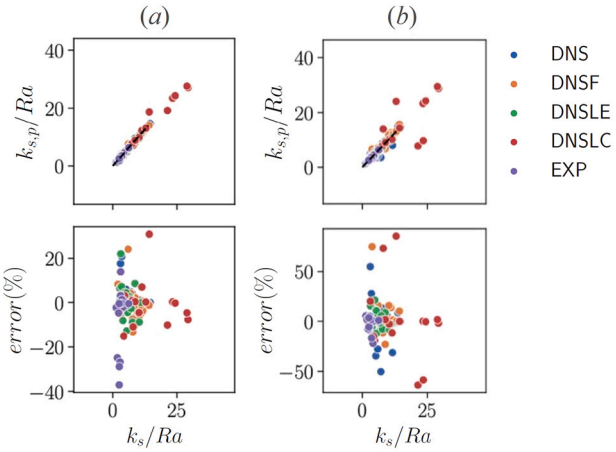

Fig. 3. Scatter plots of true k_s, predicted $k_{s,p}$ and their relative error calculated from the data-driven k_s models with different input features for (a) $\mathbf{X} = [S_x, k_{\text{rms}}, S_k, K_u, E_x, P_o]$ and (b) $\mathbf{X} = [k_{\text{rms}}, S_k, K_u, E_x, P_o]$, respectively. As shown in Table A.5, "DNS" and "DNSF" denote the DNS data of Jouybari et al. [18] and Forooghi et al. [21], "DNSLC" and "DNSLE" denote the DNS data of our previous work [45,46], "EXP" denotes the experimental data of Flack and co-authors [47,22,23]. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

The effect of input features is then examined. Without the streamwise element spacing S_x as the input feature, the maximum error becomes approximately 85% with an average error of approximately 9.5%. The comparison in Fig. 3 shows that the error is significant for the rough surfaces with cube elements arranged in a staggered way.

As a comparison to the present model, we apply four empirical k_s models and a data-driven k_s model proposed in the literature to the 118 samples of rough surfaces in Table A.5, as shown in Appendix B. The results show that the present k_s model reduces the maximum and average errors, and is applicable to more types of rough surfaces.

3. Results

To evaluate the performance of distributed force model, the WMLES of an open channel flow with different types of roughness element is carried out and compared with the DNS results.

3.1. Flow configuration

The geometries of three cube-roughened surfaces and three ellipsoid-roughened surfaces in an open channel are plotted in Fig. 4. For the DNS and WMLES of the open channel, the dimensions are set to $L_x \times L_y \times L_z = 3.0\delta \times 1.0\delta \times 1.0\delta$ and $3.0\delta \times (1.0\delta - d) \times 1.0\delta$ ($\delta = 1.0$ is the half width of the channel), thus the effective heights are $\delta_y = \delta - k_{\text{avg}}$ and $\delta - d$, where the zero plane displacement d is the equivalent lift height of the boundary layer, which depends on the type of roughness element. The bulk Reynolds number is $Re_b = U_b \delta_y / \nu = 12207$, where $U_b = Q/(\rho L_z \times \delta_y)$ is the bulk velocity, Q is the mass flux. In the streamwise and spanwise directions, periodic boundary condition is applied. And no slip and free slip boundary conditions are applied at the bottom and the top wall of the open channel, respectively. The flow is driven by a pressure gradient uniformly applied to whole domain to maintain a constant mass flux. As shown in Fig. 4, the spacing distance of roughness elements and the angle of roughness elements relative to the incoming flow (i.e. the streamwise direction) are varied for different surfaces, which could change the near-wall flow features.

Table 2 lists the parameters for the rough surfaces in Fig. 4. For the case name in the 1st column, "C" and "E" respectively denote the cube- and ellipsoid-shaped roughness element, "28" denotes the element spacing $l = 2.8r$ in the 2nd column ($r = 0.0708\delta$ is the side length of the cube element). In the 3rd and 4th columns, "n(s)45" denotes the angle 45° rotating along the vertical-y (spanwise-z) axis of the cube element. For the ellipsoid element, "p45" denotes the angle 45° rotating along the major axis (z), Rn and Ry denote the

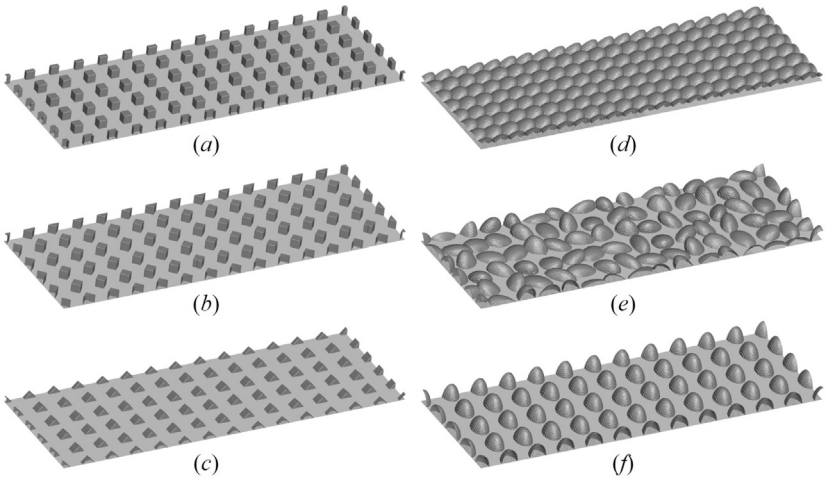

Fig. 4. Geometry of an open channel with the different cube-roughened surfaces (a-c) and ellipsoid-roughened surfaces (d-f) selected from Fig. 2: (a) C28D0A, (b) C28Dn45A, (c) C28Ds45A, (d) E20Dp45Rn, (e) E28D0Ry, (f) E35D0Rn.

Table 2
Parameters for the cube- and ellipsoid-roughened surfaces in Fig. 4, for which the DNS cases at $Re_b = 12207$ are carried out in our previous work [45,46].

Case	DNS						
	l	θ_y	θ_z	k_c/δ	k_{avg}/δ	k_s/δ	d/δ
C28D0A	2.8r	0	0	0.0708	0.0116	0.1565	0.0496
C28Dn45A	2.8r	0	45°	0.0708	0.0124	0.1567	0.0775
C28Ds45A	2.8r	45°	0	0.0497	0.006	0.2294	0.0116
E20Dp45Rn	2.0r	–	45°	0.0752	0.0561	0.0408	0.0555
E28D0Ry	2.8r	–	–	0.1023	0.0306	0.1507	0.0525
E35D0Rn	3.5r	–	0	0.1051	0.0258	0.3530	0.0540

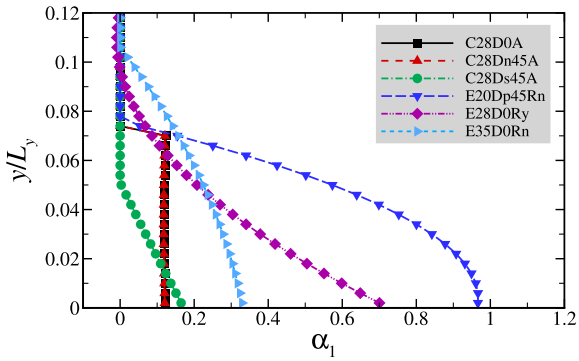

Fig. 5. Vertical distribution of area fraction for the different cube-roughened and ellipsoid-roughened surfaces in Fig. 4.

fixed and random orientations, and the semi axis lengths are respectively r, $1.4r$ and $2.0r$ in three axes. The $5^{th} \sim 8^{th}$ columns list the geometrical parameters k_c, k_{avg}, k_s and d for the rough surfaces, which are calculated from the DNSs in our previous work [45,46]. The grid resolution for the DNS cases is $512 \times 299 \times 171$. The area fractions along the $y-$axis for the different roughness elements are calculated from the DNS grids implemented with the immersed boundary (IB) method, as shown in Fig. 5. For the cube-roughened surfaces, the area fractions are constant values smaller than 0.2 at $y \le k_c$ for the "C28D0A" and "C28Dn45A" cases, and exhibit the linear growth for the "C28Ds45A" case. As for the ellipsoid-roughened surfaces, the area fractions are much larger than those from the cube-roughened surfaces and exhibit the parabolic growth along the $y-$axis. It should be noted that the vertical distribution of body force is similar to that for the area fraction because the instantaneous velocity u_i and force coefficient c_i are unchanged along the $y-$axis.

3.2. Grid resolution study

The grid resolution study is firstly performed on the distributed force model. Three groups of WMLES cases with the distributed force model, named as FR-WM-c1, FR-WM-c2 and FR-WM-c3, are carried out on the grid resolution of $75 \times 50 \times 25$, $108 \times 75 \times 36$,

Table 3
The number of grid points per k_c, the friction velocity u_τ and Reynolds number Re_τ for the DNS and three FR-WM cases of the cube- and ellipsoid-roughened surfaces at $Re_b = 12207$.

Case	DNS			FR-WM-c1			FR-WM-c2			FR-WM-c3		
	N_c	u_τ	Re_τ	N_c	u_τ	Re_τ	N_c	u_τ	Re_τ	N_c	u_τ	Re_τ
C28D0A	56	0.0911	1099	6	0.0890	1033	8	0.0930	1078	11	0.0928	1076
C28Dn45A	56	0.0952	1148	6	0.0963	1084	8	0.0939	1057	11	0.0974	1097
C28Ds45A	45	0.0945	1147	4	0.0933	1126	5	0.1005	1212	8	0.1013	1222
E20Dp45Rn	58	0.0724	834	6	0.0735	848	8	0.0736	848	12	0.0741	854
E28D0Ry	69	0.0913	1080	8	0.0931	1076	11	0.0961	1111	17	0.0955	1104
E35D0Rn	70	0.1131	1345	8	0.1137	1312	11	0.1126	1300	17	0.1136	1312

$150 \times 100 \times 50$, respectively. Table 3 lists the number of grid points per k_c (N_c), the friction velocity u_τ and Reynolds number Re_τ ($= u_\tau \delta_y / \nu$) for the DNS and three FR-WM cases of the cube- and ellipsoid-roughened surfaces in Fig. 4. We can observe that the friction velocity and Reynolds number calculated from the three FR-WM cases are close to those from DNS for all the typical roughness elements. The largest relative error of u_τ between the FR-WM-c3 and DNS, 7.2%, appears at the surface of C28Ds45A (Fig. 4c), in which the mean roughness height k_{avg} is the smallest but the equivalent sandgrain roughness height k_s is the largest in the three cube-roughened surfaces.

Fig. 6 shows the vertical profiles of mean streamwise velocity versus $(y-d)^+ = (y-d)u_\tau/\nu$ obtained from the DNS and three FR-WM cases of the open channel flows, which are normalized by the u_τ of DNS. Following standard practice [2,48,49], the logarithmic law for outer layer is adjusted by using the zero-plane displacement d. Such adjustment is only applied to the DNS profiles, as the computational domain starts from d in the wall-normal direction in the WMLES cases. In the figure, the vertical dashed lines denote the upper edge of the distributed force layer $y = k_c$. As for the FR-WM cases with different grid resolutions, the velocity profiles coincide well with each other in the outer region of $(y-d)^+ \geq 40$, and they are consistent with the DNS results for all the cube- and ellipsoid-roughened channel flows. These results demonstrate that 6∼8 points per k_c are able to accurately predict the mean velocity.

Then the second-order flow statistics are examined to investigate the grid resolution effect. Here, the comparison of vertical profiles of the primary Reynolds stress and velocity fluctuations versus $(y-d)/\delta$ at different directions between the DNS and three FR-WM cases are shown in Figs. 7 and 8 for the surface C28D0A and E20Dp45Rn, respectively. In the prediction of primary Reynolds stress $\langle u'v' \rangle^+$ and vertical velocity fluctuation $\langle v'v' \rangle^+$, the FR-WM-c2 case agrees well with the FR-WM-c3 case, while the FR-WM-c1 case exhibits slight underestimation for both the typical cube- and ellipsoid-roughened cases. As for the streamwise and spanwise velocity fluctuations ($\langle u'u' \rangle^+$, $\langle w'w' \rangle^+$), the profiles of FR-WM case gradually approach the DNS profiles with the increasing grid resolution. Compared to the FR-WM-c3 case, the FR-WM-c2 case exhibits significant underestimation at $0.1 \leq y/\delta \leq 0.6$ for the C28D0A surface (Fig. 7b, d), and slight underestimation at $0.1 \leq y/\delta \leq 0.4$ for the E20Dp45Rn surface (Fig. 8b, d). This is because the cubic roughness element has a sharper upper boundary and could generate stronger wake turbulence, for which the grid resolution has a more significant effect on the flow prediction. In total, the FR-WM-c2 case with 75 points per channel half height (8∼11 points per k_c) are required to basically predict the second-order flow statistics of cube- and ellipsoid-roughened channel flows. The similar evaluation of first- and second-order flow statistics for the distributed force model can be found in the work of Durbin and Varghese [25,49].

3.3. Evaluation of the distributed force model

To examine the performance of the distributed force model, the predicting results of FR-WM-c3 case (labeled as "FR-WM") are presented for the open channel flows with different roughness elements at $Re_b = 12207$. Fig. 9 compares the vertical profiles of mean streamwise velocity between the DNS and FR-WM. For the cube-roughened surfaces (Fig. 9a), the DNS velocity profiles remain approximately the same in the outer region of $(y-d)^+ \geq 40$ and exhibit some deviation in the near-wall region when rotating the cubic roughness elements by 45° along the vertical axis (C28Dn45A vs. C28D0A). When rotating the cubic roughness elements by 45° along the spanwise axis (C28D0A vs. C28Ds45A), the velocity profile of C28Ds45A is significantly decreased in the outer region, but close to that of C28D0A in the near-wall region. For the ellipsoid-roughened surfaces (Fig. 9b), the arrangement and spacing of the roughness elements bring about different magnitudes of drag on the fluid flow, which can be observed by the DNS value of k_s. The surface with the more closely arranged roughness elements has the smaller k_s, and the normalized velocity profile is also higher. After using the distributed force model, the velocity profiles coincide well with those from the DNS in the outer region of $(y-d)^+ \geq 40$ for the channel flows with different roughness elements. As for the near-wall region ($5 \leq (y-d)^+ < 40$), the FR-WM basically predicts the velocity profile for the surfaces of C28Dn45A and C28Ds45A, and overpredicts the velocity profile for the other surfaces.

Subsequently, Figs. 10∼11 compare the vertical profiles of second-order turbulence statistics between the DNS and FR-WM of the cube- and ellipsoid-roughened channel flows, respectively. In the near-wall flow region where the distributed force is exerted ($y \leq k_c$), the significant deviations of the profiles can be observed between the DNS and FR-WM, especially for the velocity fluctuation at the streamwise and spanwise directions. In the outer region of the flow field, the FR-WM accurately predicts the vertical profiles of different turbulence statistics for the channel flows with ellipsoid-roughened surfaces (Fig. 11). As for the cube-roughened surfaces in Fig. 10, the FR-WM accurately predicts the vertical profiles of turbulence statistics for the C28Dn45A case (c∼d), while the deviations between the DNS and FR-WM gradually increase for the C28D0A (a∼b) and C28Ds45A (e∼f) cases, especially for the primary Reynolds stress $\langle u'v' \rangle^+$ and streamwise velocity fluctuation $\langle u'u' \rangle^+$.

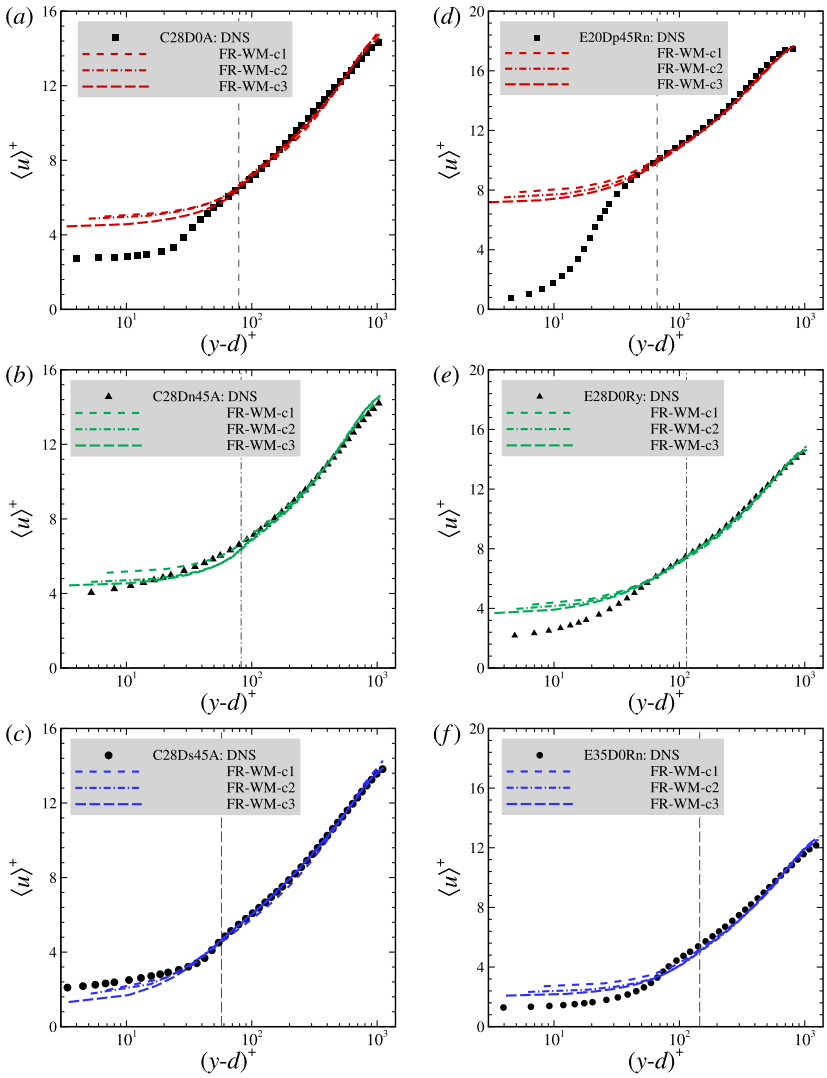

Fig. 6. Comparison of vertical profiles of mean streamwise velocity $\langle u \rangle^+$ between the DNS and three FR-WM cases of the open channel flows: (a) C28D0A, (b) C28Dn45A, (c) C28Ds45A, (d) E20Dp45Rn, (e) E28D0Ry, (f) E35D0Rn. The vertical dashed line denotes the location of k_c, which is the same in the following figures.

Table 4
The number of grid points per k_c, the friction velocity and Reynolds number for the DNS and FR-WM cases of the cube- and ellipsoid-roughened surfaces at $Re_b = 24414$.

Case	DNS					FR-WM		
	k_s	d	N_c	u_τ	Re_τ	N_c	u_τ	Re_τ
C28D0A	0.0992	0.0864	92	0.0890	2172	11	0.0877	1956
C28Dn45A	0.2173	0.0452	92	0.0902	2202	11	0.0977	2278
C28Ds45A	0.1338	0.0700	80	0.0910	2221	8	0.0965	2192
E20Dp45Rn	0.0325	0.0889	96	0.0740	1807	12	0.0727	1618
E28D0Ry	0.1535	0.0541	94	0.0887	2166	17	0.0973	2247
E35D0Rn	0.3547	0.0557	95	0.1039	2536	17	0.1188	2739

3.4. Application to the higher Reynolds number cases

In this section, the distributed force model is applied to the rough-wall channel flows at a higher Reynolds number $Re_b = 24414$. The grid resolution of DNS increases to $768 \times 399 \times 256$ for the former four cases, and $768 \times 299 \times 256$ for the latter two cases in Fig. 4. As for the FR-WM, the grid resolution of $150 \times 100 \times 50$ is the same as that for the cases at $Re_b = 12207$. Table 4 lists the number N_c of grid points per k_c, the friction velocity u_τ and Reynolds number Re_τ from the DNS and FR-WM cases of the different cube- and ellipsoid-roughened surfaces. The largest relative error of u_τ between the FR-WM and DNS, 14.3%, appears at the surface

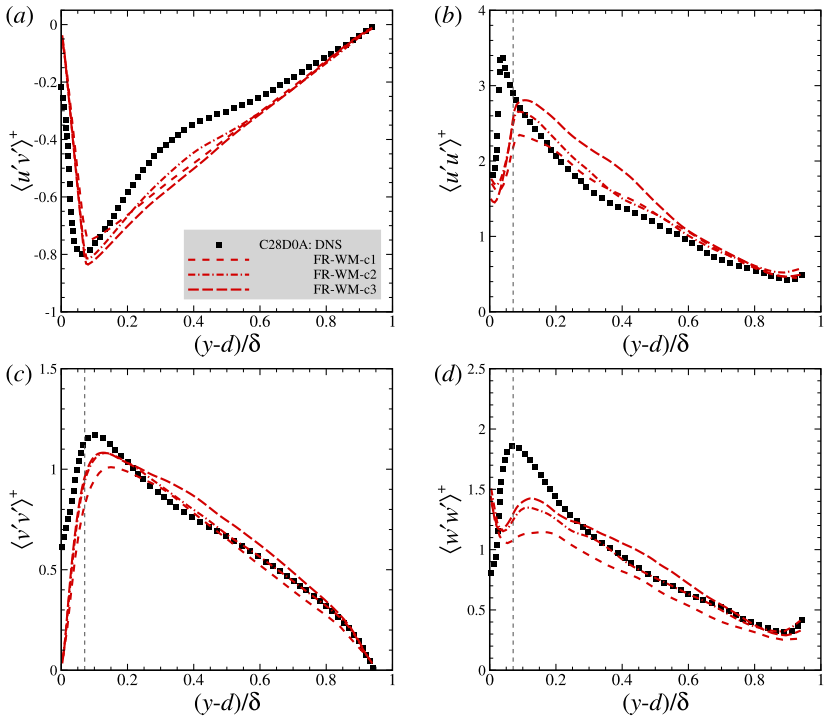

Fig. 7. Comparison of vertical profiles of (a) the primary Reynolds stress $\langle u'v' \rangle^+$ and (b~d) velocity fluctuations at streamwise, vertical and spanwise directions ($\langle u'u' \rangle^+$, $\langle v'v' \rangle^+$ and $\langle w'w' \rangle^+$) between the DNS and three FR-WM cases for the cube-roughened channel flows with C28D0A.

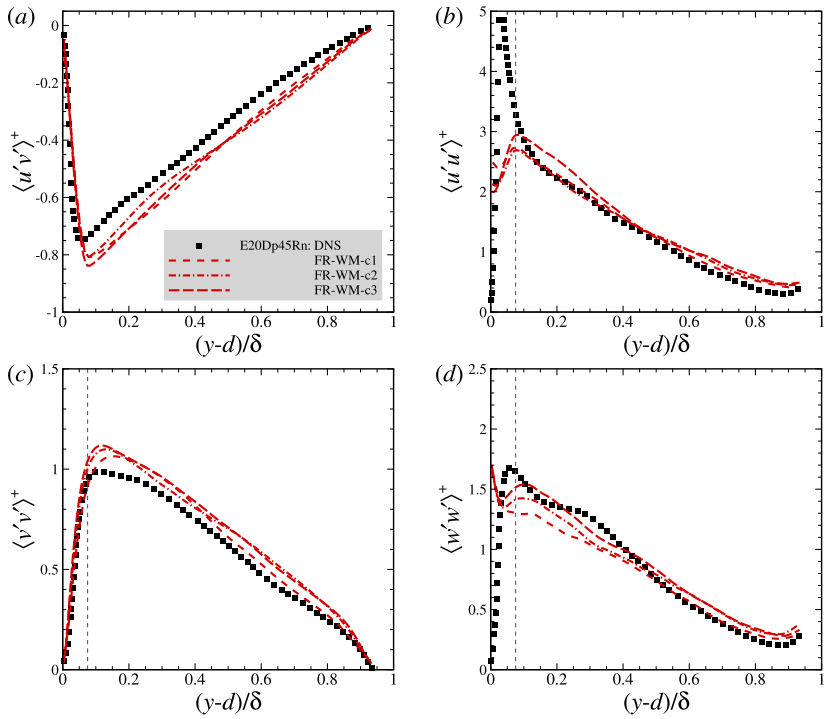

Fig. 8. Comparison of vertical profiles of (a) the primary Reynolds stress $\langle u'v' \rangle^+$ and (b~d) velocity fluctuations at streamwise, vertical and spanwise directions ($\langle u'u' \rangle^+$, $\langle v'v' \rangle^+$ and $\langle w'w' \rangle^+$) between the DNS and three FR-WM cases for the ellipsoid-roughened channel flows with E20Dp45Rn.

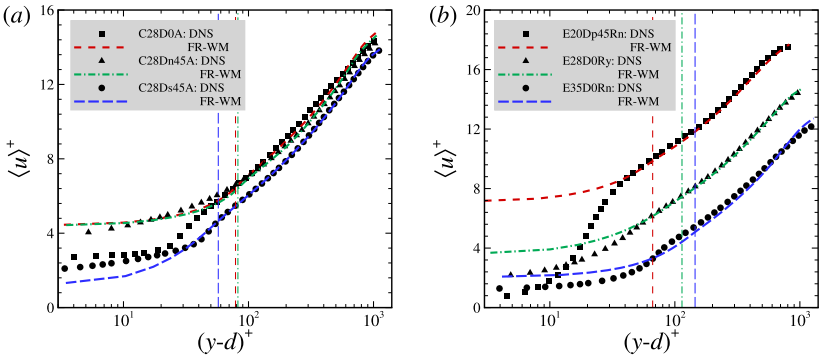

Fig. 9. Comparison of vertical profiles of mean streamwise velocity between the DNS and FR-WM of the open channel flows with (a) cube-roughened surfaces and (b) ellipsoid-roughened surfaces at $Re_b = 12207$.

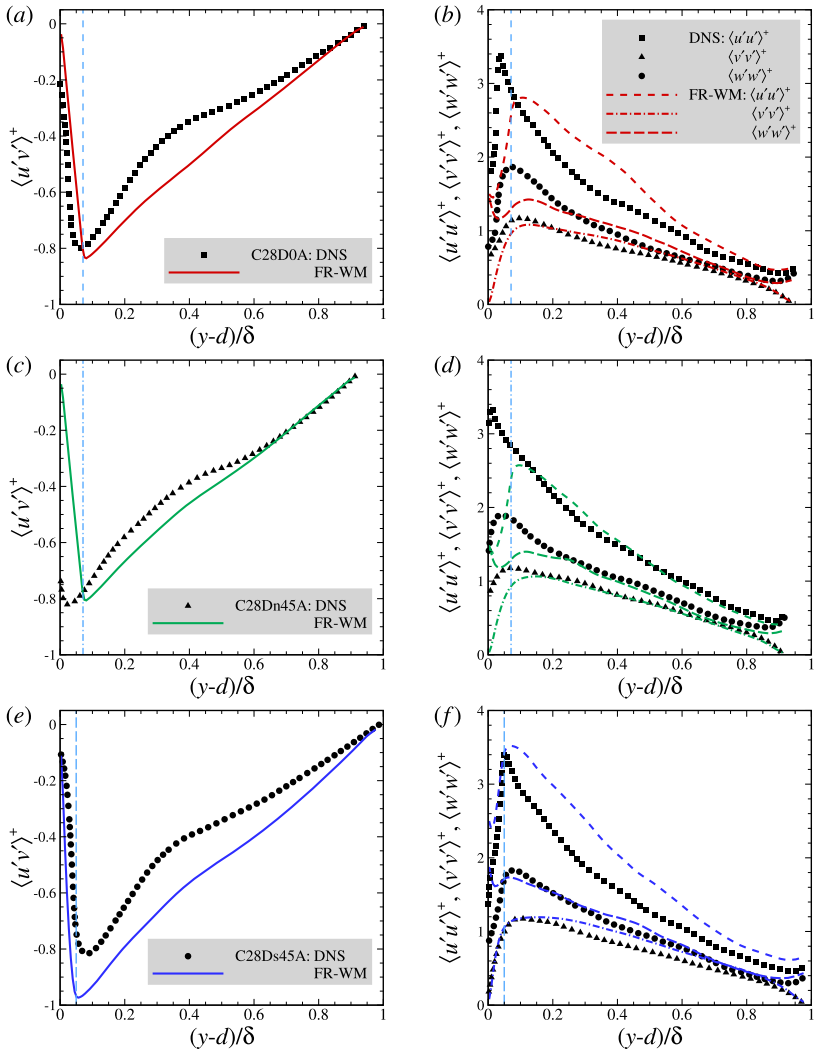

Fig. 10. Comparison of vertical profiles of the primary Reynolds stress and velocity fluctuation at streamwise, vertical and spanwise directions between the DNS and FR-WM of the cube-roughened channel flows at $Re_b = 12207$: (a)(b) C28D0A, (c)(d) C28Dn45A, (e)(f) C28Ds45A.

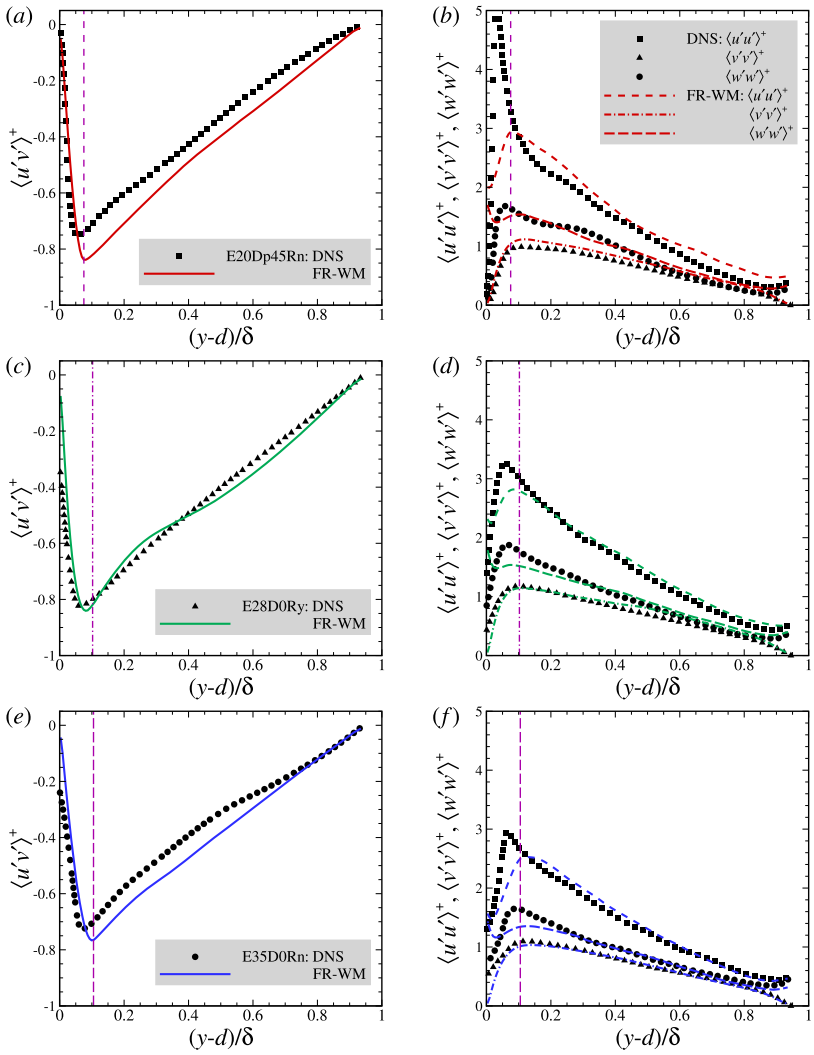

Fig. 11. Comparison of vertical profiles of the primary Reynolds stress and velocity fluctuation at streamwise, vertical and spanwise directions between the DNS and FR-WM of the ellipsoid-roughened channel flows at $Re_b = 12207$: (a)(b) E20Dp45Rn, (c)(d) E28D0Ry, (e)(f) E35D0Rn.

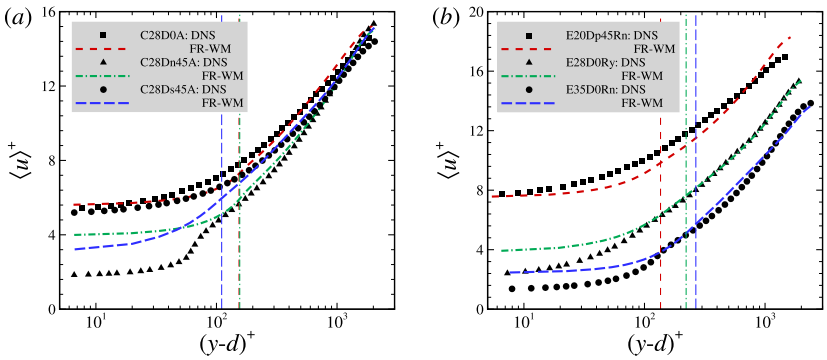

Fig. 12. Comparison of vertical profiles of mean streamwise velocity between the DNS and FR-WM of the open channel flows with (a) cube-roughened surfaces and (b) ellipsoid-roughened surfaces at $Re_b = 24414$.

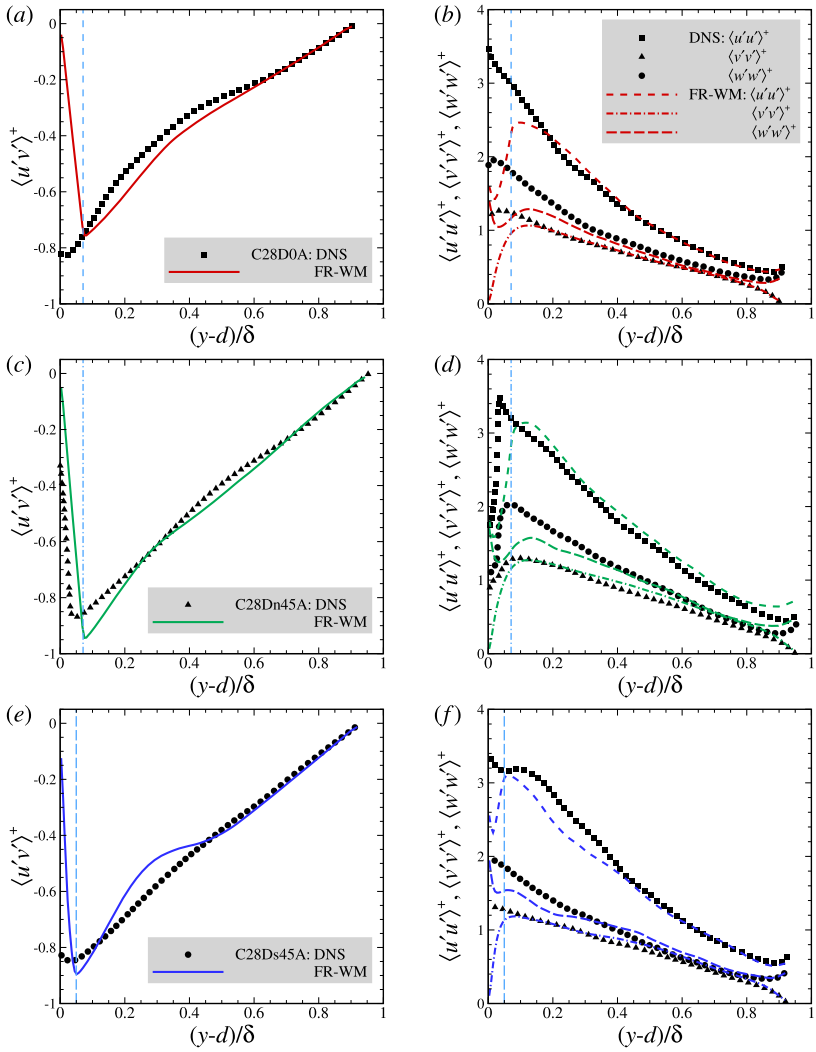

Fig. 13. Comparison of vertical profiles of the primary Reynolds stress and velocity fluctuation at streamwise, vertical and spanwise directions between the DNS and FR-WM of the cube-roughened channel flows at $Re_b = 24414$: (a)(b) C28D0A, (c)(d) C28Dn45A, (e)(f) C28Ds45A.

of E35D0Rn (Fig. 4f), in which the mean roughness height k_{avg} is the smallest but the equivalent sandgrain roughness height k_s is the largest in the three ellipsoid-roughened surfaces.

Fig. 12 compares the vertical profiles of mean streamwise velocity between the DNS and FR-WM. The velocity profiles from the FR-WM coincide well with those from the DNS in the outer region of $(y-d)^+ \geq 100$ for the channel flows with different roughness elements. As for the near-wall region ($5 \leq (y-d)^+ < 100$), the FR-WM only accurately predicts the velocity profile for the surfaces of C28D0A and E20Dp45Rn. Figs. 13~14 compare the vertical profiles of second-order turbulence statistics between the DNS and FR-WM of the cube- and ellipsoid-roughened channel flows, respectively. For the cube-roughened surfaces, the FR-WM accurately predicts the vertical profiles of primary Reynolds stress and different velocity fluctuations in the outer region. As for the ellipsoid-roughened surfaces, the vertical profiles of different turbulence statistics in the outer region are accurately predicted by the FR-WM in the E35D0Rn cases. For the E20Dp45Rn case, the streamwise and spanwise velocity fluctuations are underestimated by the FR-WM. In the E28D0Ry case, the primary Reynolds stress and streamwise velocity fluctuation are slightly overestimated.

4. Conclusions

Numerical simulations of rough-wall turbulence continue to be a difficult task because of the large variety of rough surfaces. The key for roughness-modeled simulations is to properly model the effect of a rough wall on the outer flow.

In this work, we propose a data-driven distributed force model for wall-modeled large-eddy simulation of rough-wall turbulence. In the proposed model, a body force is distributed in the near-wall roughness layer to model the way a rough wall impacts the outer flow, in which the wall-normal distribution is specified using the area fraction of the solid, and the force amplitude is computed using the local tangential velocity and the equivalent sandgrain roughness height k_s. To enable the application of the model to a

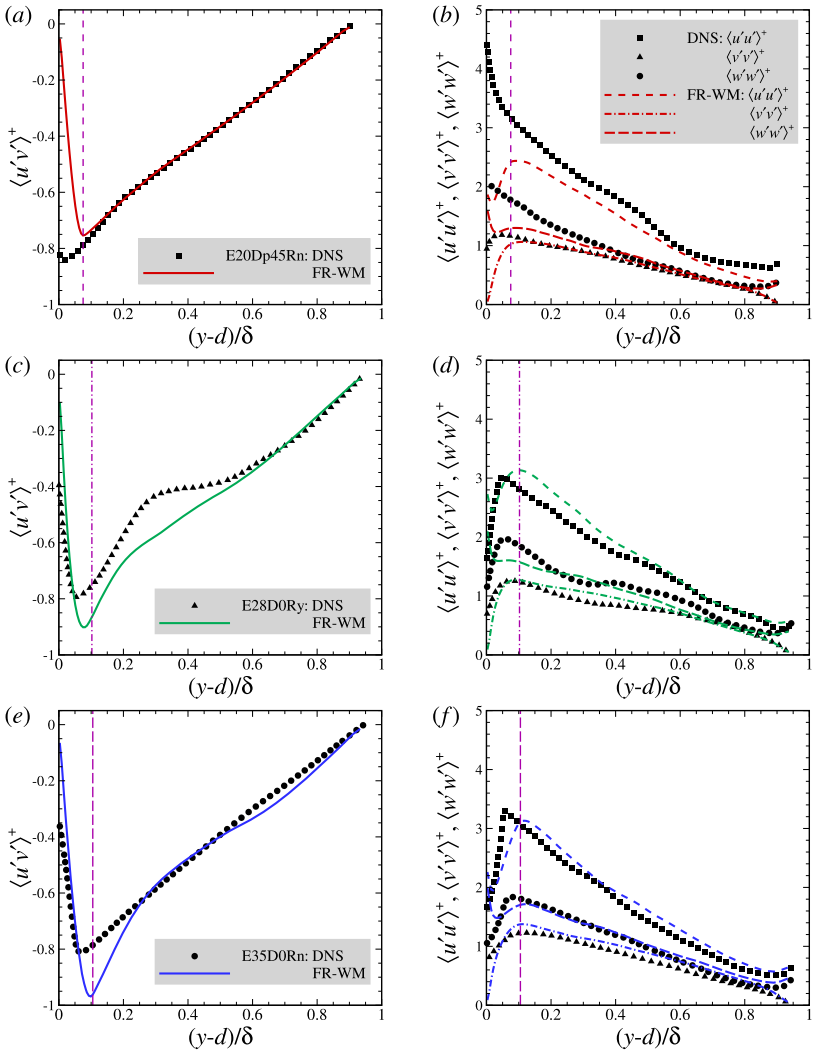

Fig. 14. Comparison of vertical profiles of the primary Reynolds stress and velocity fluctuation at streamwise, vertical and spanwise directions between the DNS and FR-WM of the ellipsoid-roughened channel flows at $Re_b = 24414$: (a)(b) E20Dp45Rn, (c)(d) E28D0Ry, (e)(f) E35D0Rn.

large variety of rough surfaces, the k_s is computed using a DNN model trained using data from experiments and roughness-resolved numerical simulations. The DNN model for k_s employs six geometrical parameters of the rough surface as the input features, which include S_x the averaged streamwise element spacing, k_{rms}, S_k, and K_u the root-mean-square, skewness, flatness of roughness height fluctuations, respectively, E_x the effective slope of roughness in the streamwise direction, and P_o the surface porosity. The maximum error of the DNN model for k_s is less than 40%, and its mean error is less than 5%.

The data-driven distributed force model for wall-modeled large-eddy simulation is tested using turbulent channel flows for three cube-roughened and three ellipsoid-roughened surfaces. The grid resolution study showed that 6~8 points per k_c (corresponding to the grid of $75 \times 50 \times 25$) is enough to accurately predict the mean velocity, and 8~11 points per k_c (corresponding to the grid of $108 \times 75 \times 36$) is required by the model to predict the second-order flow statistics. The tests on the six rough surfaces at $Re_b = 12207$ showed that for the mean velocity profile, the proposed model coincides well with the DNS result in the outer region of $(y-d)^+ \geq 40$ for all roughness elements. In the near-wall region ($5 \leq (y-d)^+ < 40$), the model accurately predicts the mean velocity profile for the surfaces of C28Dn45A and C28Ds45A, while overpredicts for the other surfaces. As for the second-order flow statistics, the model accurately predicts the profiles in the outer region for the three ellipsoid-roughened cases and the C28Dn45A case, with the discrepancies between the DNS and model predictions slightly larger for the C28D0A and C28Ds45A cases for the primary Reynolds stress and the standard deviation of the streamwise velocity fluctuations. Finally, the distributed force model is applied to the cases at a higher Reynolds number $Re_b = 24414$, with the obtained performance similar to that for $Re_b = 12207$.

Limitations of the present work include: 1) the generalizability of the data-driven k_s model. Although more than 100 surfaces were employed to construct the model, much more data are needed to ensure it has a good generalizability; 2) the assumption of statistically

homogeneous of roughness elements on a rough surface. For a surface with heterogeneously distributed roughness elements, how to account for the distribution of the surface geometric statistics is not considered in the proposed model.

CRediT authorship contribution statement

Zhideng Zhou: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Shilong Li:** Validation, Software, Data curation. **Guowei He:** Writing – review & editing, Supervision. **Xiaolei Yang:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Training dataset for the FNN model of roughness height

Table A.5 lists the dataset for learning the data-driven model for k_s. There are a total of 118 samples of rough surfaces, in which the 44 samples are the dataset used by the model of Jouybari et al. [18] (the DNS samples of Jouybari et al. [18] labeled as "DNS", the experimental samples of Flack and co-authors [47,22,23] labeled as "EXP"), 32 samples from the DNS of Forooghi et al. [21] (labeled as "DNSF") and 42 samples from the DNS of our previous work [45] (labeled as "DNSLE" and "DNSLC", where "E" and "C" denote the ellipsoid- and cube-shaped roughness element, respectively).

Table A.6 lists the maximum and minimum values of different geometrical parameters. The parameter ranges to develop the present data-driven model are $-1.44 < S_k < 2.85$ and $0.126 < E_x < 2.846$.

Appendix B. Referenced k_s model

Various models that map topographical features of the rough surface to k_s were reviewed by Flack and Chung [15]. In this appendix, four empirical formulae and a data-driven k_s model are applied to the 118 samples of rough surfaces shown in Table A.5, with their expressions given as follows:

1) Forooghi et al.'s model [21]:

$$k_s/k_t = 1.07(0.67S_k^2 + 0.93S_k + 1.3)(1 - e^{-3.5E_x}), \tag{B.1}$$

where k_t is the mean peak-to-trough height. In their work, 38 DNS cases of open channel flows with cylinder-like roughness elements in both regular and irregular arrangements (labeled as "DNSF") are employed.

2) Kuwata and Kawaguchi's model [14]:

$$k_s/k_{\mathrm{rms}} = 4.0(1 + 0.17S_k)^4, \tag{B.2}$$

which was proposed by fitting the relation between the statistical moments and k_s from DNS of open channel flows over random hemispheres.

3) Flack et al.'s model [23]:

$$k_s/k_{\mathrm{rms}} = \begin{cases} 2.48(1 + S_k)^{2.24}, \\ 2.11, \\ 2.73(2 + S_k)^{-0.45}, \end{cases} \tag{B.3}$$

which was proposed based on experimental results (labeled as "EXP"), from hydraulically-smooth to fully-rough.

4) Jouybari et al.'s model [18]:

$$k_s/R_a = 5.312 - 1.172(k_{\mathrm{rms}}/R_a)^{4.264} + 0.05I_x - 1.283|I_x|^{8.393} + \cdots, \tag{B.4}$$

Table A.5
Geometrical parameters and the equivalent sandgrain roughness height k_s for the employed 118 samples of rough surfaces, where S_x, k_{avg}, k_{rms}, R_a and k_s are normalized by the height δ of the channel.

Case	S_x	k_{avg}	k_{rms}	R_a	P_o	E_x	S_k	K_u	k_s	Label
C04-r6-reg-inc1	0.0642	0.0322	0.0219	0.0194	0.5971	0.5949	-0.1671	1.6007	0.0644	DNS
C05-r6-reg-inc2	0.0877	0.0376	0.0326	0.0298	0.6535	0.9158	0.1093	1.4355	0.1244	DNS
C06-r6-reg-inc3	0.0642	0.0311	0.0218	0.0193	0.5989	0.5876	-0.0874	1.5898	0.0589	DNS
C07-r4-rnd-inc1	0.086	0.0248	0.0223	0.019	0.7736	0.5114	0.56	2.2435	0.1362	DNS
C08-r4-rnd-inc2	0.1161	0.0273	0.0298	0.0254	0.8191	0.8612	0.8702	2.6269	0.3223	DNS
C09-r4-rnd-inc3	0.0828	0.025	0.0214	0.0181	0.7529	0.5168	0.5141	2.2923	0.1311	DNS
C10-r6-rnd-inc1	0.125	0.0256	0.0298	0.0253	0.8351	0.4979	0.9671	2.8736	0.2689	DNS
C11-r6-rnd-inc2	0.1724	0.0333	0.0437	0.0367	0.8416	0.7579	1.1501	3.1762	0.5364	DNS
C12-r6-rnd-inc3	0.1273	0.0318	0.0315	0.0266	0.7837	0.5076	0.7575	2.6415	0.2717	DNS
C14-r4-reg-por2	0.0592	0.0176	0.0217	0.0197	0.7763	0.613	0.7078	1.8403	0.1406	DNS
C15-r4-reg-por3	0.0592	0.0096	0.0182	0.0143	0.877	0.3337	1.6456	4.0941	0.1571	DNS
C16-r6-reg-por1	0.0893	0.0506	0.0304	0.0262	0.5289	1.137	-0.5377	1.8731	0.0767	DNS
C17-r6-reg-por2	0.0893	0.022	0.031	0.0274	0.8012	0.5372	0.9822	2.3078	0.2598	DNS
C18-r6-reg-por3	0.0893	0.0126	0.0259	0.0195	0.8857	0.3068	1.8486	4.8394	0.2465	DNS
C19-r4-rnd-por1	0.1119	0.0269	0.0207	0.0166	0.8058	0.487	0.7322	3.4216	0.1582	DNS
C20-r4-rnd-por2	0.0946	0.0127	0.0169	0.0139	0.8961	0.3108	1.3431	4.1256	0.1057	DNS
C21-r4-rnd-por3	0.0982	0.0089	0.0161	0.012	0.9292	0.2189	2.1683	7.7277	0.1027	DNS
C22-r6-rnd-por1	0.1393	0.0345	0.0287	0.0239	0.7909	0.4556	0.591	2.8296	0.2768	DNS
C23-r6-rnd-por2	0.1227	0.0173	0.0249	0.0202	0.8848	0.3046	1.4673	4.3466	0.1751	DNS
C24-r6-rnd-por3	0.1523	0.0135	0.0265	0.019	0.9263	0.2542	2.3707	8.7396	0.2603	DNS
C26-r4-reg-ES2	0.375	0.0207	0.0139	0.0126	0.5098	0.2124	-0.0712	1.5051	0.0653	DNS
C28-r6-reg-ES1	0.75	0.0304	0.021	0.019	0.5041	0.1581	-0.0308	1.4991	0.0712	DNS
C29-r6-reg-ES2	0.375	0.0311	0.021	0.0189	0.5041	0.3162	-0.0709	1.5031	0.112	DNS
C30-r6-reg-ES3	0.1667	0.0339	0.0203	0.0182	0.5045	0.9168	-0.2033	1.5426	0.064	DNS
C31-r4-rnd-SGR	0.0592	0.0253	0.0114	0.0093	0.6477	0.3704	0.3777	2.784	0.0487	DNS
C37-r6-rnd-SGR	0.0877	0.0369	0.0184	0.0149	0.6401	0.428	0.3231	2.6859	0.1088	DNS
C40-r6-rnd-RND3	0.1205	0.0635	0.0162	0.0129	0.5115	0.2803	0.0372	2.9768	0.0504	DNS
C43-SG	0.0889	0.0359	0.0171	0.0137	0.6492	0.4252	0.4758	2.9703	0.093	DNS
C45-CB	0.2085	0.0098	0.0228	0.0159	0.8781	0.2493	2.1006	5.5686	0.1499	DNS
A7088	0.13	0.0741	0.0417	0.0339	0.6767	0.8898	0.208	2.6225	0.3	DNSF
A7060	0.13	0.0729	0.0416	0.0339	0.676	0.603	0.2225	2.6215	0.2745	DNSF
A7040	0.13	0.073	0.0413	0.0336	0.6754	0.4034	0.2278	2.6462	0.229	DNSF
A7030	0.13	0.0745	0.0425	0.0345	0.669	0.3017	0.2568	2.6998	0.1895	DNSF
A7020	0.13	0.076	0.0418	0.0338	0.6623	0.2022	0.2553	2.7765	0.1069	DNSF
A3588	0.17	0.0803	0.0418	0.0341	0.6415	0.8914	0.2066	2.5669	0.3365	DNSF
A1588	0.204	0.0963	0.0409	0.0332	0.5593	0.8932	0.2123	2.5607	0.3676	DNSF
A0088	0.213	0.1001	0.0408	0.033	0.4777	0.8963	0.21	2.5761	0.388	DNSF
A0060	0.213	0.0997	0.0412	0.0333	0.4796	0.5961	0.186	2.5895	0.3727	DNSF
A0040	0.213	0.0996	0.041	0.0332	0.4803	0.3965	0.2026	2.5808	0.3197	DNSF
A0030	0.213	0.0994	0.041	0.0331	0.4808	0.2958	0.203	2.5966	0.2622	DNSF
A0020	0.213	0.0922	0.0441	0.0357	0.5186	0.176	0.1791	2.5156	0.1344	DNSF
B7088	0.116	0.0851	0.0411	0.033	0.5998	0.9041	-0.3248	2.6245	0.1729	DNSF
B7060	0.116	0.0854	0.0411	0.033	0.5575	0.598	-0.3568	2.6013	0.1584	DNSF
B7040	0.116	0.0848	0.0419	0.0337	0.5607	0.401	-0.3404	2.5809	0.1277	DNSF
B7030	0.116	0.0865	0.0417	0.0335	0.5271	0.2963	-0.3993	2.5967	0.1036	DNSF
B7020	0.116	0.0841	0.043	0.0346	0.5228	0.1964	-0.3407	2.489	0.0688	DNSF
B3588	0.156	0.0932	0.0413	0.0336	0.5557	0.8895	-0.3328	2.57	0.2028	DNSF
B1588	0.215	0.1272	0.0395	0.0322	0.4374	0.8873	-0.3241	2.6478	0.2137	DNSF
B0088	0.228	0.1318	0.0393	0.0321	0.3325	0.8916	-0.3362	2.5946	0.2246	DNSF
C7088	0.122	0.0474	0.0425	0.0357	0.7752	0.8865	0.6605	2.6354	0.3918	DNSF
C7060	0.122	0.0474	0.0428	0.0358	0.7753	0.6109	0.6795	2.6773	0.3648	DNSF
C7040	0.122	0.0484	0.0431	0.0362	0.758	0.3932	0.6473	2.6074	0.3019	DNSF
C7030	0.122	0.0459	0.0424	0.0358	0.7528	0.2893	0.6694	2.5607	0.2434	DNSF
C7020	0.122	0.046	0.0418	0.0353	0.7518	0.1986	0.6457	2.5532	0.149	DNSF
C3588	0.15	0.048	0.043	0.036	0.7798	0.885	0.6751	2.5969	0.4258	DNSF
C1588	0.168	0.0492	0.0429	0.0357	0.7466	0.8811	0.6749	2.5952	0.4655	DNSF
C0088	0.177	0.0515	0.0428	0.0354	0.6928	0.8818	0.6711	2.6682	0.5028	DNSF
D7088	0.165	0.0534	0.0427	0.0371	0.7016	0.8997	0.2088	1.9233	0.2113	DNSF
D0088	0.145	0.0562	0.0424	0.0366	0.5886	0.8929	0.1994	1.8358	0.2867	DNSF
D0088s	0.145	0.0573	0.0417	0.0361	0.5775	0.9306	0.2209	1.8405	0.3169	DNSF
D0088a	0.145	0.0587	0.0413	0.0355	0.5735	0.9076	0.2075	1.8979	0.2152	DNSF
E20Dn30Rn	0.14	0.0393	0.0097	0.0079	0.24	0.2873	-0.8447	3.414	0.0214	DNSLE
E20Dn45Rn	0.14	0.0562	0.0149	0.012	0.2526	0.5313	-0.9521	3.7421	0.0479	DNSLE
E20Dn60Rn	0.14	0.0643	0.0213	0.017	0.3003	0.8255	-0.986	3.4997	0.0651	DNSLE
E20Dp30Rn	0.14	0.0397	0.0098	0.0079	0.2431	0.2894	-0.8552	3.425	0.022	DNSLE
E20Dp45Rn	0.14	0.0561	0.015	0.012	0.2533	0.5324	-0.9605	3.778	0.0408	DNSLE
E20Dp60Rn	0.14	0.0644	0.0213	0.017	0.2991	0.8276	-0.9861	3.4923	0.0536	DNSLE

Table A.5 (*continued*)

Case	S_x	k_{avg}	k_{rms}	R_a	P_o	E_x	S_k	K_u	k_s	Label
E28Dn30Rn	0.196	0.0287	0.0197	0.0173	0.495	0.381	-0.18	1.6431	0.0611	DNSLE
E28Dn45Rn	0.196	0.0338	0.0269	0.0242	0.5498	0.5626	0.0144	1.5022	0.1263	DNSLE
E28Dn60Rn	0.196	0.0375	0.0333	0.0303	0.5919	0.6895	0.1524	1.4566	0.1789	DNSLE
E28Dp30Rn	0.196	0.0286	0.0197	0.0173	0.4966	0.381	-0.1764	1.6397	0.0609	DNSLE
E28Dp45Rn	0.196	0.0338	0.0269	0.0242	0.5497	0.5623	0.0137	1.5024	0.1159	DNSLE
E28Dp60Rn	0.196	0.0375	0.0333	0.0303	0.5918	0.6892	0.1518	1.4563	0.1635	DNSLE
E35Dn30Rn	0.245	0.018	0.0207	0.0188	0.6827	0.2643	0.5972	1.7412	0.0876	DNSLE
E35Dn45Rn	0.245	0.0217	0.027	0.0243	0.7119	0.36	0.7459	1.9268	0.1774	DNSLE
E35Dn60Rn	0.245	0.024	0.0321	0.0287	0.7387	0.4412	0.8703	2.112	0.2514	DNSLE
E35Dp30Rn	0.245	0.0181	0.0207	0.0188	0.6825	0.2644	0.597	1.7411	0.1025	DNSLE
E35Dp45Rn	0.245	0.0217	0.027	0.0243	0.7119	0.3598	0.7453	1.9261	0.1892	DNSLE
E35Dp60Rn	0.245	0.024	0.0321	0.0287	0.7388	0.4409	0.8707	2.1129	0.2994	DNSLE
E20D0Ry	0.14	0.0516	0.0216	0.0175	0.5091	0.4405	-0.052	2.6147	0.1026	DNSLE
E20D0Rn	0.14	0.07	0.0289	0.0234	0.3344	1.1769	-0.923	2.9433	0.0762	DNSLE
E28D0Ry	0.196	0.0306	0.0263	0.0224	0.7016	0.4585	0.4487	2.2055	0.1507	DNSLE
E28D0Rn	0.196	0.0402	0.0389	0.0357	0.6175	0.8051	0.2702	1.4657	0.2104	DNSLE
E35D0Ry	0.245	0.0212	0.0276	0.0239	0.7986	0.318	0.998	2.7194	0.1746	DNSLE
E35D0Rn	0.245	0.0258	0.0366	0.0323	0.755	0.5163	0.9774	2.3011	0.353	DNSLE
C20D0A	0.1416	0.0226	0.033	0.0308	0.6802	0.5532	0.7727	1.597	0.095	DNSLC
C20D0S	0.2832	0.0225	0.033	0.0307	0.6818	0.5504	0.7806	1.6093	0.1461	DNSLC
C20Ds45A	0.1416	0.0119	0.0169	0.0146	0.7597	0.378	1.078	2.6097	0.1506	DNSLC
C20Ds45S	0.2832	0.0119	0.0169	0.0145	0.7609	0.3761	1.0855	2.6263	0.1658	DNSLC
C20Dn45A	0.1416	0.0249	0.0338	0.0323	0.6484	0.7258	0.6216	1.3864	0.1175	DNSLC
C20Dn45S	0.2832	0.0249	0.0338	0.0323	0.6485	0.7253	0.622	1.3869	0.1401	DNSLC
C28D0A	0.19824	0.0116	0.0262	0.0195	0.8355	0.2788	1.8102	4.2769	0.1565	DNSLC
C28D0S	0.39648	0.0116	0.0262	0.0194	0.8367	0.2769	1.8215	4.3177	0.2772	DNSLC
C28Ds45A	0.19824	0.006	0.0134	0.0095	0.8793	0.1901	2.152	6.2277	0.2294	DNSLC
C28Ds45S	0.39648	0.006	0.0134	0.0095	0.8801	0.1888	2.1638	6.2811	0.2212	DNSLC
C28Dn45A	0.19824	0.0124	0.027	0.0205	0.8242	0.3629	1.7034	3.9015	0.1567	DNSLC
C28Dn45S	0.39648	0.0125	0.027	0.0205	0.8241	0.3632	1.7027	3.8993	0.438	DNSLC
C35D0A	0.2478	0.0077	0.0221	0.0138	0.8906	0.1892	2.5031	7.2657	0.1221	DNSLC
C35D0S	0.4956	0.0077	0.022	0.0137	0.8916	0.1875	2.519	7.3454	0.3206	DNSLC
C35Ds45A	0.2478	0.0041	0.0113	0.0069	0.9156	0.1269	2.8294	9.759	0.2024	DNSLC
C35Ds45S	0.4956	0.004	0.0113	0.0069	0.9163	0.1258	2.8461	9.8592	0.1997	DNSLC
C35Dn45A	0.2478	0.0085	0.023	0.015	0.8798	0.2467	2.3365	6.459	0.1945	DNSLC
C35Dn45S	0.4956	0.0085	0.0231	0.015	0.8795	0.2484	2.3318	6.4373	0.3654	DNSLC
E01-16-2	0.03676	0.138	0.0204	0.0161	0.4716	0.7202	-0.7109	3.8425	0.052	EXP
E02-16-3	0.03695	0.1434	0.0205	0.0164	0.432	0.7401	-0.338	3.1589	0.05	EXP
E03-16-7	0.03525	0.1326	0.0185	0.0141	0.6375	0.6177	-1.1689	5.2921	0.058	EXP
E04-16-8	0.03688	0.1259	0.0165	0.0127	0.5791	0.587	-1.4448	5.4214	0.056	EXP
E05-16-9	0.04438	0.1123	0.0175	0.0139	0.6367	0.6362	-0.7384	3.7139	0.043	EXP
E06-16-15	0.02813	0.0809	0.0126	0.0098	0.5775	0.6209	-0.6874	3.8541	0.035	EXP
E07-18-1	0.036	0.1206	0.0264	0.0211	0.4999	0.181	0.1074	2.9405	0.053	EXP
E08-18-2	0.036	0.143	0.0318	0.0254	0.4832	0.1624	0.0934	2.967	0.034	EXP
E09-19-1	0.036	0.2041	0.0455	0.0364	0.487	0.2268	-0.0797	2.9891	0.065	EXP
E10-19-2	0.036	0.3887	0.0877	0.0702	0.4916	0.4466	-0.0651	2.9253	0.2	EXP
E11-19-3	0.036	0.4772	0.0881	0.0704	0.348	0.4336	-0.6595	3.2739	0.16	EXP
E12-19-4	0.036	0.4585	0.0885	0.0708	0.3913	0.4551	-0.3512	3.0406	0.18	EXP
E13-19-5	0.036	0.2922	0.0895	0.0715	0.6016	0.4448	0.3458	3.0505	0.245	EXP
E14-19-6	0.036	0.2022	0.0866	0.0692	0.7163	0.3909	0.8117	3.5588	0.435	EXP
E15-19-7	0.036	0.5216	0.1142	0.092	0.4621	0.5569	-0.0662	2.7937	0.23	EXP

Table A.6
The maximum and minimum values of different geometrical parameters.

	S_x	k_{avg}	k_{rms}	R_a	P_o	E_x	S_k	K_u
Max	0.5216	0.5216	0.1142	0.092	0.9292	1.1769	2.8461	9.8592
Min	0.004	0.004	0.0097	0.0069	0.24	0.1258	-1.4448	1.3864

which was proposed using a 30-degree-freedom white-box polynomial formula by fitting the data from 45 samples of rough surfaces (labeled as "DNS" and "EXP").

5) Jouybari et al.'s model [18], which is a DNN model for predicting k_s and is accessed at https://github.com/ MostafaAghaei/ Prediction-of-the-roughness-equivalent-sandgrainheight.

Fig. B.15 compares the scatter plots of real k_s, predicted $k_{s,p}$ and their relative error calculated from different models, while the maximum and average errors are listed in Table B.7. For the three empirical models in Fig. B.15(a)~(c), the scatter plots of $k_{s,p} \sim k_s$ exhibit significant deviation from the dashed line with unity slope, the maximum and average errors are more than 200% and 30%, respectively, which are much larger than the present DNN model. It should be noted that the large errors are mainly caused by the unseen rough surfaces. Taking the model of Kuwata and Kawaguchi [14] as an example, the maximum and average errors

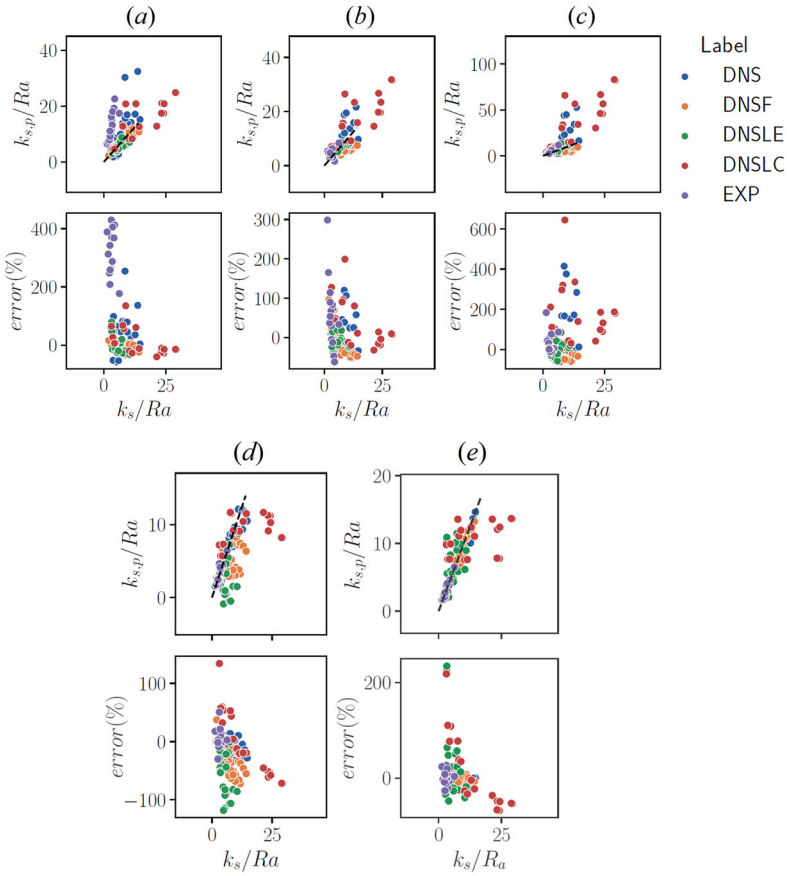

Fig. B.15. Scatter plots of true k_s, predicted $k_{s,p}$ and their relative error calculated from the empirical model proposed by (a) Forooghi et al. [21], (b) Kuwata and Kawaguchi [14], (c) Flack et al. [23], (d) Jouybari et al. [18], and (e) the DNN model proposed by Jouybari et al. [18].

Table B.7
The maximum and average errors of various models on predicting k_s for various rough surfaces.

Model for predicting k_s	$error_{\max}$ (%)	$error_{\text{avg}}$ (%)
Forooghi et al. [21]	429.4462	65.6766
Kuwata and Kawaguchi [14]	298.4505	37.3199
Flack et al. [23]	644.2211	67.2999
Jouybari et al. [18]: polynomial	134.1227	34.2316
Jouybari et al. [18]: DNN	234.2523	23.5126
Present: DNN	37.1046	4.8511

decrease from (298%, 37%) to (39%, 17%) for only the "DNSLE" samples, whose surface geometries are similar to those used in the literature [14].

As for the white-box polynomial model and DNN model proposed by Jouybari et al. [18], the scatter plots of $k_{s,p} \sim k_s$ are closer to the dashed line, but the errors of some scatter points in Fig. B.15(d)∼(e) are still larger than 50%, which cause the average error of 34% and 24%. The scatter points with large errors appear mainly to the "DNSLE" and "DNSLC" samples, which have not been seen in the literature. When the test dataset is only composed of "DNS", "DNSF" and "EXP" samples, the maximum and average errors reduce to (72%, 24%) for the polynomial model, and (27%, 7.6%) for the DNN model of Jouybari et al. [18].

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